

Student Performance Predictor

Early Prediction of Student Performance

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PROJECT OVERVIEW

Quick Summary

Predicts student final exam performance using historical academic data.

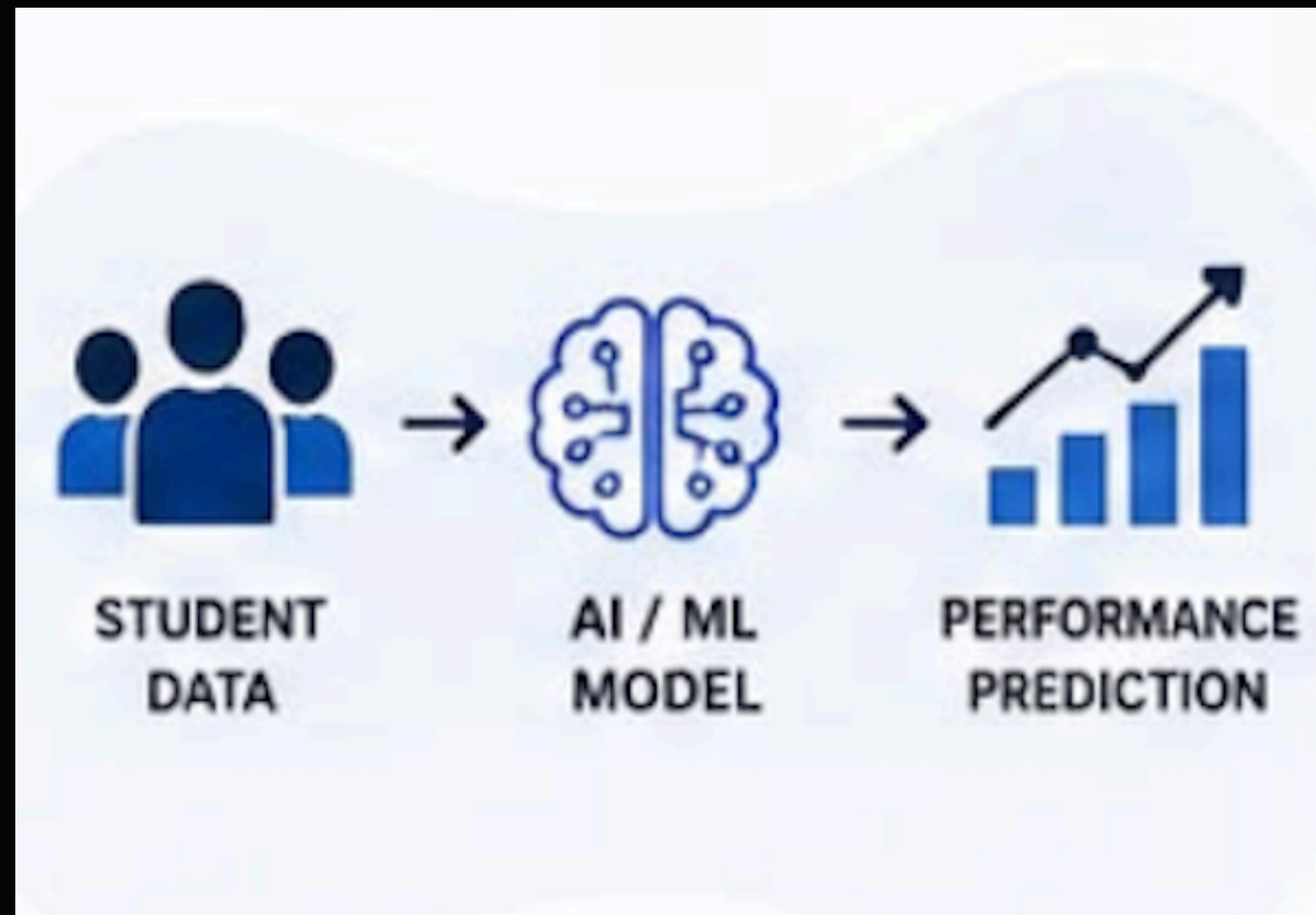
Helps identify learning patterns and supports early academic intervention.

Key Highlights

- **Student performance prediction**
 - **Data visualization**
 - **Multiple ML model comparison**
 - **Performance evaluation**
 - **Early academic insights**

Objective: Predict student performance before final examinations.

Problem Statement



- Teachers often identify weak students only after exams.
- Manual performance tracking is slow and time-consuming.
- Early warning signs are difficult to detect.
- Data can help predict performance before final exams.

Objective →

Build a machine learning model that predict student performance using historical academic data.

Dataset & Exploratory Data Analysis

Dataset Overview

1000 student records

7 input features

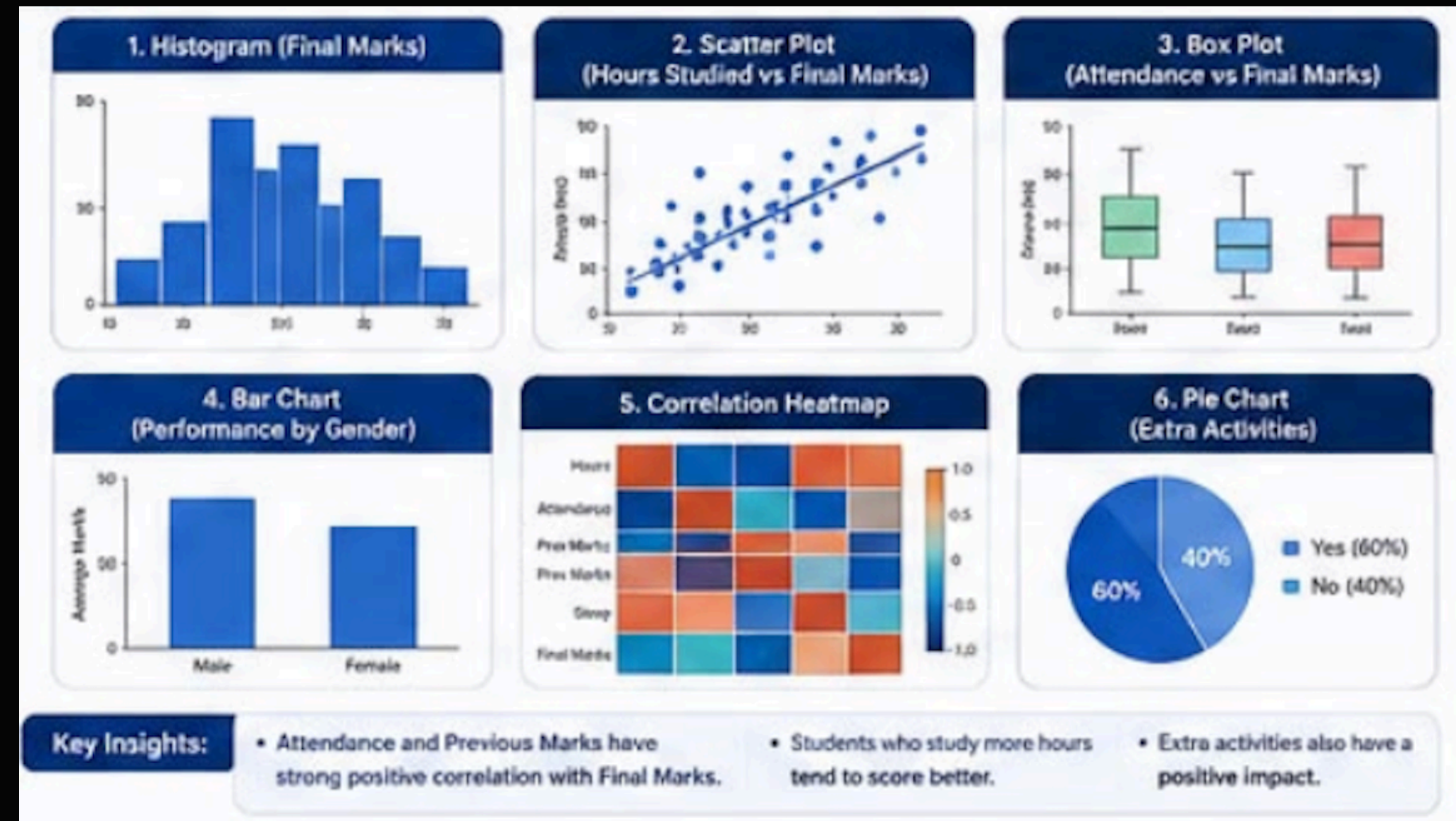
Target: Final Exam Marks

Data Preparation

Removed missing values

Cleaned and organized the dataset

Prepared data for model training



Prediction Demo →

1 → Studied.

2 → Attendance

3 → Previous Marks

4 → Sleep Hours & Assignments

Predicted Result →

Predicted Final Marks: 88/100

Performance Level: Excellent

The screenshot displays a web-based prediction tool. On the left, a form titled 'Input Student Details' contains six input fields: 'Hours Studied' (6), 'Attendance (%)' (92), 'Previous Marks' (80), 'Sleep Hours' (7), 'Assignments Completed (%)' (85), and 'Extra Activities (Yes = 1, No = 0)' (1). A blue 'Predict Marks' button is at the bottom. A blue arrow points to the right, leading to a results panel titled 'Predicted Final Marks'. This panel features a gold medal icon, the score '88 / 100', and the text 'Performance Level Excellent'.

Input Student Details	
Hours Studied	6
Attendance (%)	92
Previous Marks	80
Sleep Hours	7
Assignments Completed (%)	85
Extra Activities (Yes = 1, No = 0)	1

Predicted Final Marks


88 / 100
Performance Level
Excellent

The prediction helps teachers identify students who may need academic support before final examinations.

THANK
YOU.